

Satellite cloud data over France

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Data survey, 7 October 2026

Two kinds of satellites

Orbit	Images of the same area	Pixel size	Examples
Geostationary	Every 2.5–15 min [*]	0.6–5 km	MSG (SEVIRI), MTG (FCI)
Polar orbiting	1–2 per day, or every few days for finer images	10 m–1 km	Sentinel-2, Landsat, MODIS, VIIRS

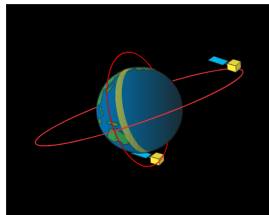
Geostationary

Keeps the same region in view.

Polar orbiting

Passes near the poles.

Sees a new strip as Earth turns.



[Open GIF](#)

^{*}2.5 min: planned MTG rapid scan. Animation: [NASA Space Place](#).

Full disc or rapid scan: the whole view or only Europe

edge of view: 81°N



Full disc

All the Earth the satellite sees, about 40% of the globe

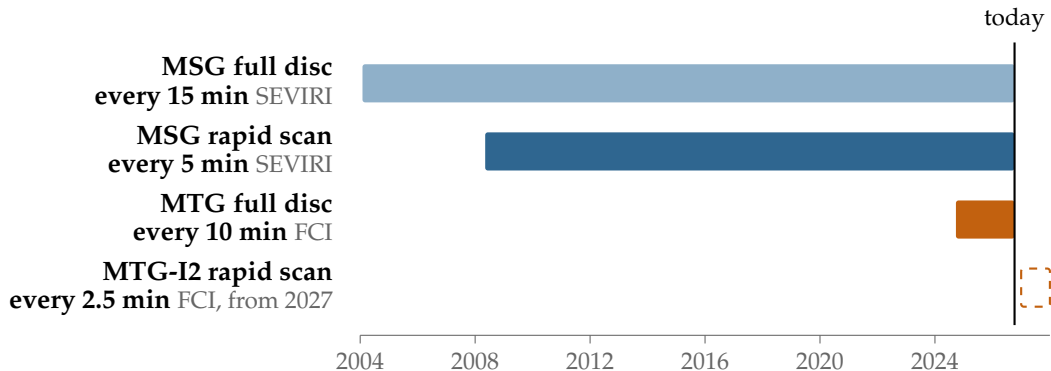
Rapid scan

Only the northern third: Europe and North Africa. Smaller area, scanned more often: with SEVIRI (MSG) every 5 min instead of 15.

the Earth as the satellite sees it

France is in both

Which geostationary satellite covers which years



Full disc is always the slower one: 15 against 5 min (MSG), 10 against 2.5 (MTG)

Three kinds of channels, three questions

The 12 SEVIRI channels (MTG has 16)

pixel at 45 °N, east–west × north–south

Visible

daytime

Where are the clouds?

VIS0.6 0.56–0.71 μm , VIS0.8 0.74–0.88 μm

$3.2 \times 5.2 \text{ km}$

Visible, HRV

daytime

The same, with pixels 3 times finer

broadband 0.4–1.1 μm , about 5 times wider than VIS0.6

$1.1 \times 1.7 \text{ km}$

Near-infrared

daytime

Ice or water?

NIR1.6 1.50–1.78 μm

$3.2 \times 5.2 \text{ km}$

Thermal infrared

day and night

How cold, so how high?

8 channels, 3.5–14.4 μm : IR3.9, WV6.2, WV7.3, IR8.7, IR9.7, IR10.8, IR12.0, IR13.4

$3.2 \times 5.2 \text{ km}$

HRV: pixels 3 times finer in both directions, and a visible band 5 times wider.

12 channels, each providing its own measurement.

Reflectance (%)

Visible and near-infrared

How much sunlight is reflected

Brightness temperature (K)

Thermal infrared

How warm or cold it appears

No correction for the Sun's angle: low Sun makes the same ground look darker.

	Every	Years	Whole archive
"Images": what the instrument measures			
MSG rapid scan	5 min	May 2008 →	101 TB
MSG, copy by Open Climate Fix	5 min	2008–2022	62 TB
MTG full disc	10 min	Sep 2024 →	107 TB
Products: one answer per pixel, by EUMETSAT			
MSG cloud mask	5 min	Oct 2008 →	0.15 TB
MSG cloud-top height	15 min	Jun 2005 →	0.24 TB
MTG cloud mask	10 min	Jan 2025 →	0.45 TB
MTG cloud-top height, cloud type	10 min	Dec 2025 →	3.9 TB

Start dates: first product in the EUMETSAT Data Store, checked 6 Oct 2026. MTG images start in Sep 2024, pre-operational; operational since 4 Dec 2024.

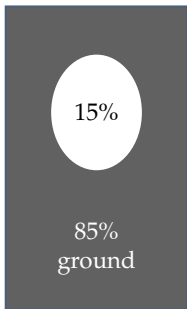
Two ways to get the same SEVIRI images

	EUMETSAT	Open Climate Fix
Archive 2008–2022	≈ 81 TB	≈ 62 TB
One download	58 MB Whole Europe, one scan	≈ 1.4 MB Regional block, one hour
France, one day	≈ 8.4 GB	≈ 0.24 GB

Open Climate Fix lets us download only the blocks covering France: about 1/35 of the download volume.

Mainland France, 25 July 2020, 08:00–20:00 local. Daily totals include HRV; 1.4 MB is a standard-channel block.

One pixel, two answers: image or cloud mask



a 3×5 km pixel
white: cloud footprint

Image

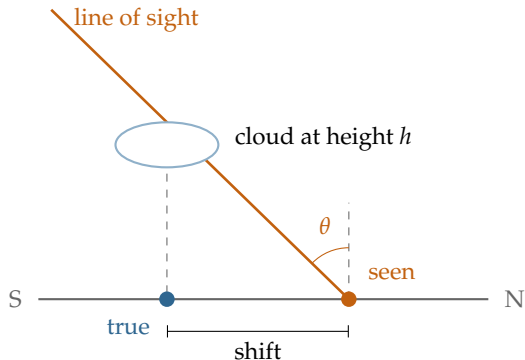
The small cumulus raises reflectance from 10% to 19%.

Cloud mask one label for the whole pixel

A small cumulus may not trigger a “cloudy” label.

Our thermal study targets small cumuli, which the standard mask can miss. We therefore want to use images, especially HRV.

Parallax: clouds appear north of where they are



$$\text{shift} = h \tan \theta \approx 1.2\text{--}1.5 \times h$$

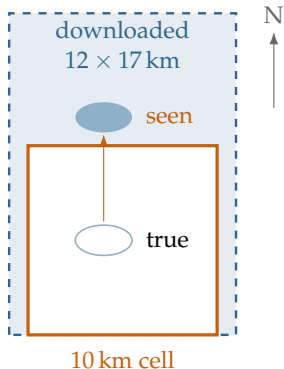
θ : angle of the line of sight from vertical,
50–57° over France

Cloud at 2.5 km 3–4 km north

Cloud at 4 km 5–6 km north

A cloud above a 10 km cell can show up in the next cell to the north.

Parallax correction: small cumuli can lack height estimates

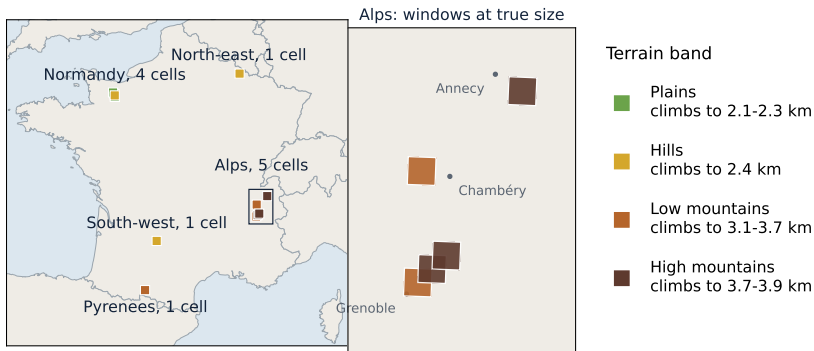


- 1 Satellite cloud-top height is the intended input for parallax correction
- 2 **Small cumuli may only partly fill a pixel.** Cloud and ground signals mix; MTG's height product excludes fractional clouds.
- 3 Climb heights are a possible fallback?

The day's highest altitude reached in a climb need not mark cloud base and does not measure cloud top.

EUMETSAT MTG cloud-top temperature and height: fractional clouds excluded. Satpy: correction uses a supplied height.

Our study: twelve 10 km windows

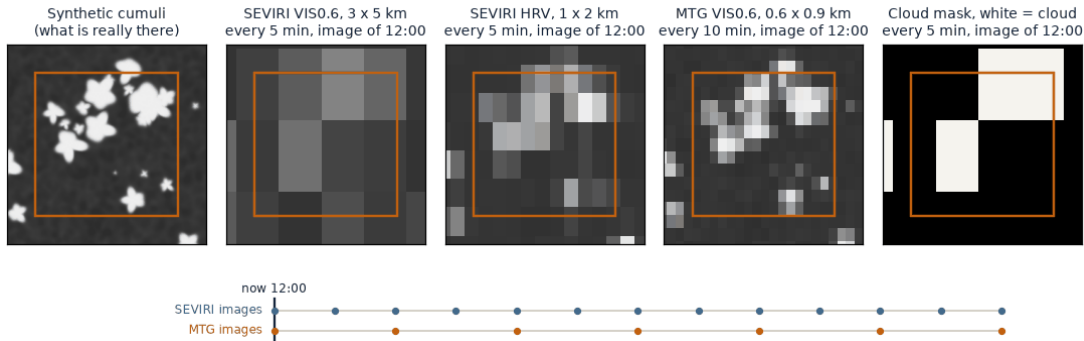


Every 5 min? 15min?, 2008–2026? 2008–2023?, 08:00–20:00 ?

Terrain band: highest IGN DEM elevation inside each 5 km cell.

What each dataset would show in a 10 km cell, over one hour

Illustration of what to expect: synthetic clouds, not real satellite data



[Open GIF](#)

Illustration, not data. At 3×5 km, SEVIRI is too coarse for our analyses; HRV shows the larger cumuli, MTG single clouds. Each panel changes only when its satellite takes an image.

Three quarters of our flight-days fall in the cheap years



2008–2022

SEVIRI copy: cheap

2023–2024

SEVIRI files: expensive

2025–2026

MTG: feasible

SEVIRI runs to today; only its cheap copy stops in 2022.

	Kept	Downloaded
1 SEVIRI images 2008–2022, OCF copy	5.4 GB	420 GB
2 SEVIRI cloud mask 2008–2026	0.16 GB	85 GB
3 MTG images 2024–2026, 3 strips	1.1 GB	3.3 TB
4 Optional: SEVIRI cloud-top height	0.13 GB	149 GB
Total	≈ 7 GB	≈ 4.0 TB

Left out: SEVIRI images for 2023–2026, 11.6 TB to download for 1.2 GB kept.

EUMETSAT Data Store: product descriptions, counts and file sizes	queried 6 Oct 2026
Open Climate Fix SEVIRI rapid-scan Zarr, public bucket	block sizes measured 6 Oct 2026
Open Climate Fix Satip: channel scaling constants	satip/constants.py
MTG FCI Level 1 Product User Guide	40 strips per scan
Meteosat-12 declared operational	4 Dec 2024
MTG-I2 launch	27 Aug 2026
Bley and Deneke (2013), HRV cloud mask	EUMETSAT mask and sub-pixel cumuli
Deneke et al. (2021), HRV implementation	Sect. 3.5 and data availability
Deneke et al. (2021), public example data	case studies, not our full archive
WMO OSCAR: SEVIRI	sampling and field of view
satpy parallax correction	needs cloud-top height